

HACK-A-DOODLE-DO

The targets of hackers have shifted. Instead of typical desktops or laptops, hackers will now be focussing more on smartphones. The fact is that breaking into a smartphone gives them access to pretty much all about the user and his or her banking details. The smartphone has emerged as the most popular personal computing device around the world, and a single device incorporates what was previously a camera, a personal media player, a gaming console, a personal digital assistant, as well as a mobile phone. It was only recently that Lebanese group, Dark Caracal used Android spyware to steal hundreds of gigabytes of sensitive data from more than 21 countries. Types of stolen data included audio recordings, text messages, call records, documents, photos, contact information, secure messaging client content, account data and enterprise intellectual property. It was all done via modified APKs of popular secure messaging apps such as Signal and WhatsApp. Expect more such attacks in 2018.

We've seen the power of IoT botnets with Mirai, a botnet that used a host of compromised IoT devices. Since then, we've seen many more IoT devices come onto the internet, be it in the form of smart speakers, connected headphones, wearables or the regular smartphones, laptops and more. Since Mirai surfaced, quite a few variants have shown up - some of which cast a much wider and/or deeper net. These variants can be built in ways that fulfil multiple purposes - mine/hack cryptocurrencies or carry out DDoS attacks. The scale of the attacks are unprecedented, with major content delivery networks and dedicated DDoS prevention tools being unable to keep up with the intense firepower of Mirai based DDoS attacks.

Such operations can be really taxing on the hacked devices and can cause temporary to permanent physical damage. Check out the all-encom-

Credit: Kaspersky Lab



A deformed smartphone after a period of cryptocurrency mining

passing malware Loapi (<http://dgit.in/loapijoat>) that can subscribe users to paid services, send SMS messages to any number, generate traffic and make money from showing advertisements, use the computing power of a device to mine cryptocurrencies, as well as perform a variety of actions on the internet on behalf of the user/device. It causes infected devices to show

visible signs of stress damage within a couple of days, to the point where the battery expands enough to deform the device. Cryptocurrency miners can put enough load on smartphones and IoT devices, to the extent of physically damaging them.

As more parts of our lives are getting connected, there's more loopholes for hackers to exploit since not all companies or users are incentivised or motivated to secure their systems. A recent hack in a major Las Vegas casino was traced to a connected fish tank on the premises. Hackers in 2018 are not as focussed on innovating as they are on scaling up. So, it's more likely than ever before that you will be targeted as a part of a large scale attack on any of the several connected fronts in your life. As long as we adapt the major security protocols that we are actually supposed to follow in the first place, and are careful about the sources we install our applications from, we should be secure from most hacking threats in 2018. ■

TEACH A MAN TO PHISH AND YOU FEED HIM FOR A LIFETIME

Phishing involves impersonating a trusted individual or company, to trick a user into compromising their sensitive data. For example, to gain your banking credentials or email login details, a malicious attacker might send you a link appearing to originate from the company and directing you to a web page that looks identical to the official website. Google and Amazon are among the most spoofed companies.

More sophisticated attacks can say be personalised to each individual in an organisation, and bait the users with the promise of a chance to win a free gift. If everyone on a mailing list has been sent a fishy email, be especially wary of follow up emails supposedly from cybersecurity personnel, providing mitigation measures against the initial round of messages. Those could be phishing attacks as well. If the initial round of phishing emails was in the news, follow up attacks are more likely to succeed. Even suspicious users will see enough material in the news to convince them of the situation after looking up the claims of the attacker on the internet.

Phishing remains one of the most effective and lucrative methods for cybercriminals. Emerging technologies promise to allow the attackers to craft even more convincing and targeted emails. One of the biggest additions to the toolkits of hackers, is AI. AI will allow cybercriminals to mimic various